News Publisher Reliability from Article Collections: A Publisher-Disjoint Evaluation

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract. Reliability ratings are assigned to news publishers, yet computational approaches often formulate the task as article-level classification, inheriting publisher labels at the article level and, in some cases, allowing the same publishers to appear in both training and test data. This mismatch creates a risk of publisher leakage and leaves unclear whether the reported performance generalizes to previously unseen outlets. We reformulate the task as publisher-level reliability inference from multiple articles. Our two-stage framework first derives a reliability prediction from each article and then aggregates these predictions to assign the publisher one of five ordinal reliability levels. We evaluate four transformer-based models using five-fold publisher-disjoint cross-validation on 19,476 English-language political news articles from 439 publishers annotated with NewsGuard scores. The best models achieve approximately 0.60 exact accuracy at the article level and 0.69 after aggregation at the publisher level. When predictions adjacent to the reference level are accepted, accuracy reaches 0.84 and 0.88, respectively, indicating that most errors remain local on the ordinal scale. Larger decoder-based models provide no consistent advantage over encoder-based models in this setting. We additionally examine how errors vary across publishers' political leanings, highlighting the importance of complementing aggregate performance with group-level error analysis. These results show that aggregating evidence across articles supports reliability estimation for unseen publishers while providing a stricter protocol for evaluating cross-publisher generalization.

Keywords: News source reliability · Transformer-based classification · Cross-publisher generalization · Political bias analysis

1 Introduction

Assessing the reliability of news publishers has become increasingly important for limiting the spread of misleading or low-quality information. Professional organizations such as NewsGuard, the Global Disinformation Index, and Media Bias Fact Check evaluate news outlets according to journalistic standards including transparency, editorial accountability, factual accuracy, and correction practices, and despite methodological differences they exhibit substantial convergence [11,19]. These assessments are performed at the level of the publisher

rather than of individual articles. Trained analysts produce them by hand, so the news outlets they never reach stay unrated.

A large body of computational work instead focuses on article-level classification [1,14,21,24]. However, assessing a publisher across multiple articles can provide a more robust estimate of its reliability than evaluating individual articles in isolation. This approach better reflects the structure of professional assessment, in which reliability is understood as a persistent property of a source: even highly reliable outlets occasionally publish lower-quality content, while unreliable publishers may produce individual articles that appear plausible.

Moreover, article-level evaluation is vulnerable to publisher leakage. When article-level reliability labels are inherited from publisher-level annotations, assessing whether a model can generalize to articles from previously unseen outlets requires publisher-disjoint training and test sets. Some existing studies do not enforce this separation [1,6,23], allowing articles from the same publisher to appear in both sets. Models may therefore exploit publisher-specific patterns instead of learning transferable indicators of reliability [3,26].

We address these limitations by framing reliability assessment as publisher-level inference. Our two-stage framework uses transformer-based language models to derive a publisher-reliability prediction from each article and then aggregates these predictions across articles to assess previously unseen publishers. We make four main contributions: (i) a strict publisher-disjoint evaluation protocol that prevents source leakage and measures generalization to unseen publishers; (ii) a dataset of 19,476 political news articles from 439 English-language publishers annotated with NewsGuard scores; (iii) evidence that aggregating predictions across articles improves reliability estimation, achieving 0.69 publisher-level accuracy on unseen sources; and (iv) an analysis of how prediction errors are distributed across the political spectrum. Given the potential consequences of such assessments for the outlets being evaluated, Section 6 also articulates the need motivating this work, the conditions under which it may provide benefits, and the harms it may introduce.

2 Related Work

Early work on online news quality focused on article-level classification, detecting misleading content through linguistic features or neural architectures [14], often without considering the publisher.

Publisher reliability and the signals used to estimate it A parallel line shifts the focus to the source, variously termed *reliability*, *trustworthiness* or *credibility* [8, 15,27], whose reputation is known to influence perceived information quality [9]. Computational approaches rely on textual and stylistic features [21] or on external signals such as user interactions, social networks and metadata [4,13,20,28]; journalistic rating systems are commonly used as ground truth [11,19], and recent work explores LLM-based estimation [10,18,24,29].

Datasets and cross-domain generalization Datasets with explicit publisher reliability labels remain limited [1,5,23], and recent work shows that performance degrades under distribution shifts across domains and publishers [3,6,26].

Our contribution In contrast, we focus exclusively on textual content, enforce a strict publisher-disjoint evaluation protocol and report the error disparities the resulting estimates carry.

3 Problem Definition and Methodology

3.1 Problem definition

We formulate the problem as 'News Source Reliability Inference': estimating the reliability level of a publisher from its published content. Because publishers are observed through collections of articles, we decompose it into (i) article-level reliability estimation and (ii) publisher-level inference through aggregation.

Let A be a corpus of news articles, each with a text in $\mathcal{T}$ and a publisher $p(a) \in P$, and let $A_p = \{a \in A \mid p(a) = p\}$ be the articles of p. Each publisher has a fixed reliability level $r_P(p)$ over five ordered categories $L = \{l_1, \ldots, l_5\}$ obtained by binning continuous scores (Section 4); since reliability is a publisher's property, every article inherits it, $r_A(a) = r_P(p(a))$. A political leaning $o_P(p) \in O$, with O ranging over Far Left, Slightly Left, Non labelled, Slightly Right and Far Right, is inherited in the same way. Both inheritance functions are needed because article-level annotation is unavailable; political leaning is never a model input or target, and is retained only for the post-hoc analysis of Section 5.4.

In **Stage 1** we train a classifier $C_{\text{reliability}} : \mathcal{T} \to L$ to predict the reliability of a single article from its text alone. In **Stage 2** we aggregate those predictions through $Agg : L^{|A_p|} \to L$, so that $\hat{r}_P(p) = Agg(\{C_{\text{reliability}}(a) \mid a \in A_p\})$; Section 5.5 instantiates Agg as the mode, i.e. majority voting.

Because each article inherits its publisher's label, articles from the same source are not independently labeled. A random article-level split could therefore place articles from the same publisher in both training and test sets, leaking publisher–label associations across the two partitions. Letting P_{train} and P_{test} denote the sets of publishers represented in the training and test data, respectively, we enforce

$$P_{\text{train}} \cap P_{\text{test}} = \emptyset. \tag{1}$$

All test articles thus originate from publishers that are entirely unseen during training. This protocol reflects the source-level nature of the labels and prevents models from improving test performance by memorizing publisher-label associations. For the same reason, we do not mask source identifiers: because every test publisher is absent from the training data, its identifier is also unseen and cannot serve as a key to a memorized publisher-label association.

3.2 Evaluation Metrics and Baselines

We use the same metrics for both article-level and source-level evaluations.

Standard and Macro Metrics We report Accuracy as the ratio of correct predictions. Because class frequencies differ moderately (l_4 is about 2.6 times larger than l_5), we prefer Macro-averaged Precision, Recall and F1, which treat all five classes equally.

Tolerant Metrics Given the ordinal nature of our labels, standard accuracy may be overly punitive for minor deviations. We therefore adopt *Tolerant Accuracy* [17], which assigns partial credit to predictions falling into adjacent categories. Let N be the total number of samples and c_{ij} the count of instances with true label i predicted as j:

$$\text{Accuracy}_{\text{tolerant}} = \frac{1}{N} \sum_{i \in L} \sum_{j \in L} w_{ij} \cdot c_{ij}, \qquad w_{ij} = \begin{cases} 1 & \text{if } |i-j| = 0 \\ \alpha & \text{if } |i-j| = 1 \\ 0 & \text{if } |i-j| > 1 \end{cases} \tag{2}$$

The parameter $\alpha \in [0, 1]$ controls the tolerance level. We report results for $\alpha = 0.5$ (half credit for adjacent errors) and $\alpha = 1.0$ (full credit, effectively treating adjacent classes as correct).

Baselines We compare against two baselines that ignore the article text, and report both of them on every metric. The **Majority Baseline** assigns every instance to the most frequent class observed in the training set, which is the best any constant predictor can do on accuracy; it scores zero Recall and F1 on every other class, so its macro figures are low by construction. The **Uniform Random Baseline** draws each of the five reliability levels with equal probability. Its expected accuracy, Macro Precision and Macro Recall are all 0.20, while its expected Macro F1 is 0.19, slightly lower because the classes are unevenly represented. Being stochastic, it is reported as the mean over 1,000 seeded draws.

3.3 Bias Assessment and Statistical Analysis

Beyond aggregate performance, we investigate *how* and *why* models fail.

Error Analysis We compute the Error Rate per level, $(1 - \text{Recall}_l)$. To identify systematic over- or under-estimation, we analyze the Mean Error Direction (MED) over the set E of misclassified samples:

$$\text{MED} = \frac{1}{|E|} \sum_{i \in E} (\hat{y}_i - y_i) \tag{3}$$

where $|E|$ is the number of such samples, and $\hat{y}_i$ and y_i represent the predicted and true reliability levels (mapped to their ordinal values $1, \ldots, 5$), respectively. Positive values indicate a tendency to overestimate reliability, while negative values indicate a tendency to underestimate reliability. Correctly classified items are excluded from the average, so the MED describes the direction of the errors and not how often they occur.

Statistical Significance of Political Bias To determine whether classification errors depend on the publisher's political leaning, we conduct a Pearson's χ^2 test of independence. We test the null hypothesis (H_0) that error frequency is independent of political orientation.

To quantify the magnitude of this effect independent of sample size, we compute Cramér's V:

$$V = \sqrt{\frac{\chi^2}{N \cdot \min(r-1, c-1)}} \tag{4}$$

where N is the total number of samples, while r and c represent the number of rows and columns of the contingency table that crosses political orientation with classification outcome. Here $r = 5$ orientations and $c = 2$ outcomes, correct and incorrect, so $\min(r-1, c-1) = 1$ and V reduces to $\sqrt{\chi^2/N}$. The tables themselves, one per model, are reported in the Supplementary Material [2].

4 Dataset and Labeling

Ground Truth: NewsGuard The publisher-level reliability function $r_P : P \to L$ is derived from NewsGuard,[1] a professional media rating system in which trained journalists evaluate online news outlets. Each publisher $p \in P$ is assessed against nine widely adopted journalistic criteria, including the use of deceptive headlines, transparency of ownership and financing, and correction practices.

The evaluation produces a continuous reliability score in $[0, 100]$, measuring a publisher's adherence to professional journalistic standards.[2] The proprietary ratings were licensed for use to the authors of this publication.

Labels and Selection of Publishers We discretize the continuous score into five ordinal categories $L = \{l_1, \ldots, l_5\}$ following NewsGuard's own thresholds. Table 1 reports the mapping, the initial number of publishers, the actual scraped domains, and the final number of articles per label.

Label	Score Range	Initial Dom.	Scraped Dom.	Final Art.
l_1	0–39 (Max Caution)	~~1,211~~ → 175	152	4,871
l_2	40–59 (Caution)	127	69	3,016
l_3	60–74 (Credible w/ Exc.)	124	95	4,620
l_4	75–99 (Generally Credible)	175	88	5,041
l_5	100 (High Credibility)	68	35	1,928

Table 1. Reliability labels and NewsGuard score ranges, with the number of domains after filtering, successfully scraped domains, and collected articles.

[1] https://www.newsguardtech.com/
[2] https://www.newsguardtech.com/ratings/rating-process-criteria/

We restrict P to publishers tagged as *Political News and Commentary*, publishing mainly in English and without a paywall. To prevent the corpus from being dominated by highly unreliable publishers, we cap l_1 at 175 publishers, matching l_4; the initial set contains $|P| = 669$ news outlets.

Article Collection Let A denote the set of collected news articles. For each publisher $p \in P$, we scrape articles from its homepage using the `newspaper3k` library[3], collecting at most 200 articles per outlet to ensure a balanced representation across publishers. The scraper is configured to retrieve English-language content only and to prevent duplicates. Scraping did not succeed for all outlets, due to anti-scraping measures and timeouts, so the final publisher count is lower than the original selection (669 vs 439). We retain only articles of at least 100 words, excluding short announcements and advertisements. This procedure yields an initial corpus of 25,822 articles.

Data Curation and Labeling To preserve the original linguistic characteristics of the articles, we apply no text normalization or cleaning. Some domains nonetheless returned localized content, which we filter out with the `langdetect` library[4]. This process removes 6,346 articles and leaves 19,476 English-language articles in the dataset.

Political Leaning Annotation NewsGuard also provides a political leaning annotation for a subset of publishers, inherited at the article level as $o_A(a) = o_P(p(a))$. Reliability levels are unevenly represented across political categories, a structural property of the media ecosystem that must be considered when interpreting model performance and error patterns in the analyses that follow.

5 Experiments

This section presents the experiments and results.

5.1 Experimental Setup

Following the validation protocol described in Section 3 (Equation 1), we use a 5-fold cross-validation scheme with strict separation of publishers between training and test sets. This assesses generalization to articles from previously unseen publishers and approximates a realistic *in-the-wild* deployment. Folds are stratified on publisher reliability, so that each preserves the same reliability distribution.

We select four transformer-based models spanning encoder baselines and decoder-based large language models. **BERT**-base-uncased [7] establishes a lower bound comparable with a vast body of prior work [22]. **RoBERTa**-large [12]

[3] https://pypi.org/project/newspaper3k/
[4] https://pypi.org/project/langdetect/

tests whether optimized training strategies alone improve reliability detection with the architecture unchanged. **Llama 3** 8B [25] represents the shift toward decoder-only generative models, and **Mistral** 24B [16], roughly three times larger, tests whether additional capacity improves detection or whether performance saturates.

Training Strategies and Hyperparameters Encoders (BERT, RoBERTa) were *fully fine-tuned* with AdamW, a learning rate of $2e^{-5}$ and weight decay [7,12]. For the decoders (Llama, Mistral) full fine-tuning was computationally infeasible, so we adopted Low-Rank Adaptation (LoRA) with a learning rate of $2e^{-4}$ in mixed precision.

Each model was trained for 5 epochs per fold, except Mistral, which was trained for 3. This follows a preliminary convergence study (see Supplementary Material [2], which also reports the remaining optimisation and batching details) showing that validation loss minimizes at Epoch 4 for Llama, while Mistral converges within three epochs. We used early stopping on validation loss and a fixed global random seed (42). The maximum sequence length for input processing was set to 256 tokens, except for Mistral, for which it was increased to 1,024 tokens. This tests whether a longer context improves classification.

All fine-tuned models and code will be released to encourage further research. They are withheld during the review process to preserve anonymity, but will be made available in case of acceptance.

5.2 Stage 1: Article-level Reliability Prediction

We first evaluate the article-level predictors of Stage 1.

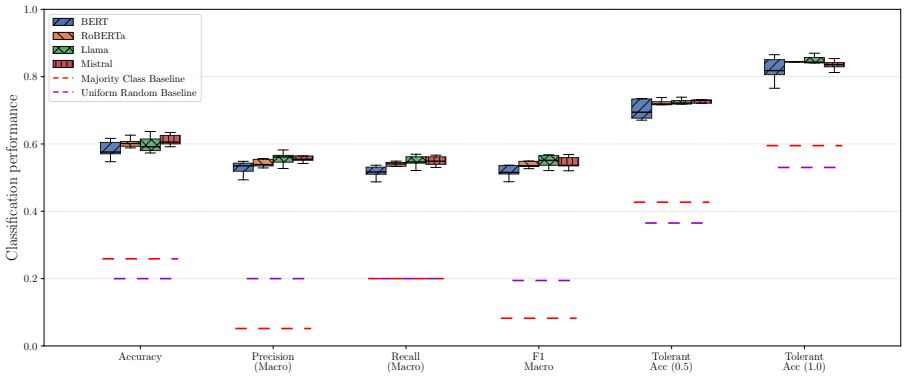


Fig. 1. Article-level metrics across the five folds. Both baselines are drawn across every metric: red for Majority Class, violet for Uniform Random. Where the two coincide, as on Macro Recall at 0.20, the dashes alternate.

Figure 1 illustrates the results, with both baselines drawn across every metric: the Majority Class Baseline (MCB) in red and the Uniform Random Baseline (RB) in violet. All models reach a standard accuracy around 0.60, more than double the MCB, with Mistral highest at 0.61. On macro-averaged performance RoBERTa leads with $F1_{macro}$ of 0.55, followed by Mistral and Llama at 0.54 and BERT at 0.52, all well above the RB of 0.19; the variance across folds is minimal (standard deviation about 0.02). Under the stricter tolerance ($\alpha = 0.5$), Mistral, RoBERTa and Llama reach approximately 0.72 against an adjusted MCB of 0.43. With $\alpha = 1.0$ RoBERTa reaches 0.84 and the others 0.82–0.83; the baseline is naturally higher here (0.60), so the gain is +0.24. Misclassifications are therefore rarely severe, typically involving adjacent labels.

5.3 Biases in Article-level Reliability Prediction

We examine whether classifiers consistently perform better in certain reliability classes than in others.

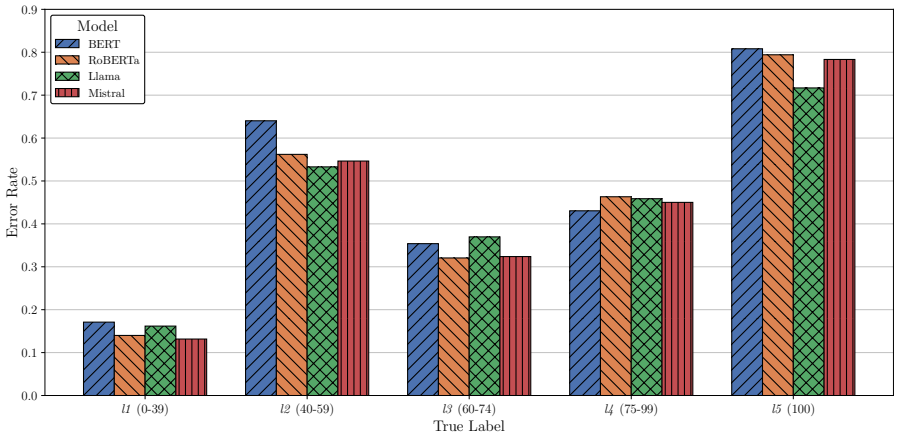


Fig. 2. Per-class error rate, $(1 - \text{Recall}_l)$.

Figure 2 highlights an asymmetry in prediction correctness. The lowest error rates are on l_1 (< 0.20), suggesting that highly unreliable content has recognizable structural features. The highest-reliability articles (l_5) are the most challenging to recognize for the model, with error rates exceeding 70%[5]. Class l_2 also proves difficult, with an error rate of over 50%, while l_3 sits in the middle at approximately 35%.

[5] Class l_5 only includes articles with a reliability score of 100, and is easily confused with l_4, which contains similarly high scores.

Model	Accuracy	χ^2	p-value	Cramér's V
BERT	58.27%	1684.13	$< 10^{-300}$	0.294
RoBERTa	60.34%	1418.31	$< 10^{-300}$	0.270
Llama	59.97%	1207.83	$< 10^{-250}$	0.249
Mistral	61.16%	1343.98	$< 10^{-300}$	0.263

Table 2. Political leaning vs. classification correctness: χ^2 test of independence with p-values, and Cramér's V as effect size.

These patterns remain consistent across all four models, suggesting that the biases are structural properties of the task rather than of a specific architecture; the Supplementary Material [2] reports the error-direction analysis and the full confusion matrices.

5.4 Influence of Political Leaning on Classification Errors

We investigate whether the models treat all sides of the political spectrum equally. Political orientation is a property of the publisher, inherited by each article it publishes (Section 3.1), which enables a post-hoc analysis of how source-level alignment relates to article-level errors.

Error rates are computed separately for each leaning, which makes them independent of class frequency.

All models exhibit systematic variation across political leanings: there are substantially higher error rates for far left and far right outlets (59–72%) compared to lower rates for slightly right and non-labelled outlets (30–34%), and intermediate error rates for slightly left articles (47–58%).

A chi-square test of independence rejects, for every model, the hypothesis that correctness is independent of political leaning (Table 2), with Cramér's V indicating a consistent small-to-moderate effect size. Leaning is a publisher-level attribute, and every article of an outlet carries the same value. We therefore repeat the test with one observation per publisher, and the association holds: χ^2 ranges from 18.8 to 44.8, $p < 10^{-3}$, and V from 0.21 to 0.32. Leaning and reliability are themselves entangled, and no analysis separates them completely: 142 of the 207 slightly right publishers sit in l_1, and no far-right publisher reaches l_5. A regression that controls for the true reliability level still leaves the two extremes with 4.0 to 6.5 times the odds of error of non-labelled outlets, but it rests on thin cells (Supplementary Material [2]). The disparity is therefore a property of the corpus and the model together. A system trained on this ground truth inherits the political structure of the dataset, and the disparity would appear in deployment whatever its mechanism.

We then focus on misclassified instances to assess whether models systematically overestimate or underestimate reliability as a function of political leaning. Figure 3 reports the Mean Error Direction (MED) for each orientation (see Section 3.3); positive values indicate overestimation of reliability, negative values underestimation.

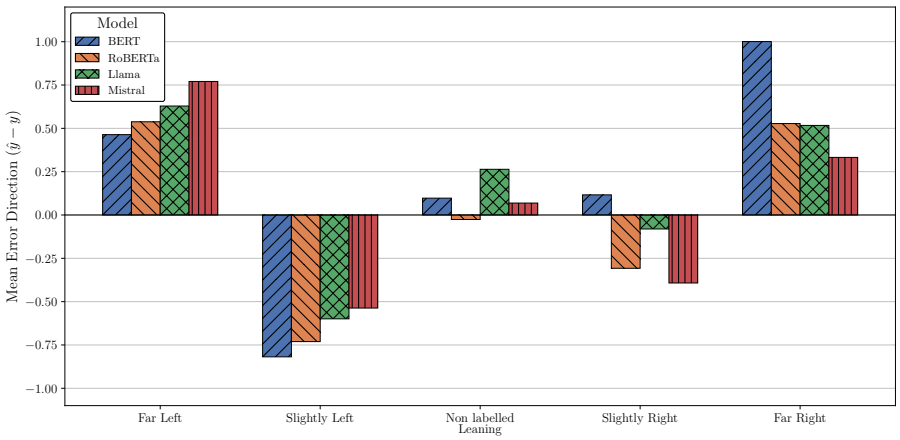


Fig. 3. Mean value and direction of misclassifications, stratified by political leaning. Positive values indicate overestimation (prediction > true label), while negative values indicate underestimation.

Clear directional patterns emerge across all four models. Far Left and Far Right articles exhibit consistent overestimation, with mean signed errors ranging from +0.33 to +1.00, approaching a full reliability level. Slightly Left articles show systematic underestimation, with average values between -0.54 and −0.82. Slightly Right and non-labelled outlets display smaller, less consistent deviations; the Supplementary Material [2] reports the corpus composition by leaning, the per-leaning error rates and the full confusion matrices stratified by political leaning.

5.5 Stage 2: News Source Reliability Inference

Publisher-level inference is the primary objective of this work; the article-level predictors are intermediate. We instantiate *Agg* as the mode of a publisher's article-level predictions, i.e. majority voting, assigning the most frequent class as $\hat{r}_P(p)$, on the hypothesis that combining predictions offsets the noise of individual classifications. Predictions are pooled across the five test folds, so each publisher's reliability is inferred only from articles the model never saw in training.

Figure 4 presents the results against the same two baselines. The majority-class reference differs across the three accuracy variants, since a constant predictor also collects partial and then full credit under tolerance. Under full tolerance the random baseline (0.52) in fact overtakes it (0.50), since a constant predictor earns credit only next to its own level.

With an Accuracy of 0.69 RoBERTa leads, followed closely by Mistral at 0.68, more than triple the Random Baseline. Considering $\text{Accuracy}_{\text{tolerant}}$ (1.0), Llama achieves the top score of 0.88, followed by RoBERTa (0.87) and Mistral (0.86), substantially outperforming the majority class baseline.

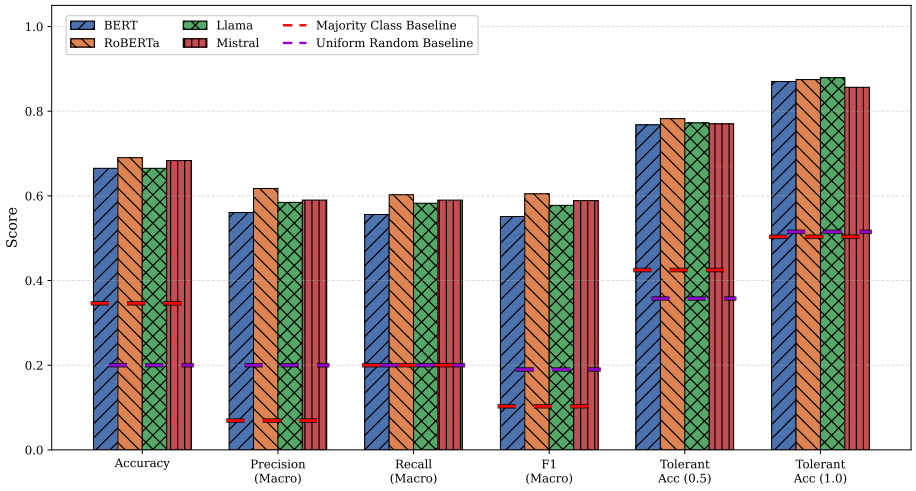


Fig. 4. Publisher-level metrics, with both baselines across every metric. Red: Majority Class Baseline; violet: Uniform Random Baseline. Where they coincide the dashes alternate.

Publisher-level errors are strictly local, concentrating between l_4 (Generally Credible) and l_5 (High Credibility); the Supplementary Material [2] reports full publisher-level confusion matrices and a detailed error analysis.

6 Theory of Change

Work on misinformation is not socially beneficial by default. We therefore state what we expect this work to change, what must hold for it to do so, and how it may cause harm.

The Need and What We Claim About It Expert ratings rely largely on manual assessment, leaving publishers outside their coverage unrated. This gap motivates automation, but expanding coverage is not itself the contribution of this paper. Instead, we change how the task is formulated and evaluated. Reliability labels apply at the publisher level, yet much computational work predicts them one article at a time, and some studies allow articles from the same outlets to appear in both training and test data. Under such protocols, a model may rely on outlet-specific cues rather than transferable indicators of reliability, so its reported performance may not generalize to unseen sources. We therefore formulate the task at the publisher level, enforce publisher-disjoint splits, and aggregate evidence across multiple articles. Under this stricter setting, our best method achieves 0.69 accuracy on unseen publishers (Section 5.5); most errors involve adjacent reliability levels, and larger models show no consistent advantage in our experiments. The protocol and the accompanying audit are released

as reusable code, so that the change we propose can be applied to corpora and label scales other than ours.

Preconditions and Their Plausibility The applicability of our approach depends on four conditions, whose plausibility varies across settings. **The reference labels must be appropriate for the intended use**: our models reproduce the NewsGuard scale and therefore inherit its editorial judgements. Our experiments assess whether these labels can be predicted, not whether the scale itself is a definitive measure of reliability. **Predictions must be interpreted at the publisher level**: because article labels are inherited from their sources, neither the supervision nor the observed article-level accuracy supports claims about the reliability of individual articles. **Sufficient textual evidence must be available**: the approach requires multiple articles per publisher, which may limit its applicability to small, recent, or infrequently publishing outlets. **Generalization must be validated in the target domain**: our evidence concerns English-language political news, and applications to other languages, domains, or time periods require further evaluation.

Negative Externalities Our experiments reveal one potential source of harm: error rates vary systematically across political-leaning groups (Section 5.4), reaching 59–72% for far-left and far-right outlets, compared with 30–34% for slightly right and non-labelled outlets. Among misclassified instances, the models also tend to overestimate reliability at both political extremes and underestimate it for slightly left-leaning outlets. These disparities should not be attributed to political leaning alone, because leaning and reliability are unevenly distributed in the corpus and cannot be fully separated by our analysis. Nevertheless, deploying the models without accounting for this uncertainty could reproduce or amplify the observed disparities.

Two additional risks concern how the predictions are presented and used. First, assigning a single level on a five-point scale may convey more precision than the models support, particularly because they struggle to distinguish l_4 from l_5 (Section 5.3). Second, once an automated score influences rankings or access to resources, publishers may adapt their writing to superficial cues learned by the model without improving the underlying editorial practices.

Scope of the Evidence Our evaluation shows how accurately the proposed approach predicts reliability labels for previously unseen publishers under a publisher-disjoint experimental protocol. It does not establish whether these predictions improve human assessment in practice, how publishers should be prioritized for expert review, or when the system should abstain because too little textual evidence is available. Answering these questions would require a deployment-oriented evaluation involving explicit decision rules and human assessors. We therefore present the approach as a potential source of decision support, not as a validated replacement for expert evaluation. The reporting that these conditions imply is released as a checklist alongside the code and the fold assignment, so that they can be applied rather than only read.

7 Discussion and Conclusion

In this study, we investigated whether the reliability of news publishers could be inferred from their articles, framing the task as a five-level ordinal classification problem aligned with NewsGuard ratings and evaluating four transformer-based models under a strict publisher-disjoint protocol.

Our findings demonstrate that language models can capture meaningful and generalizable signals of publisher reliability. Interestingly, the substantially larger Mistral 24B does not outperform smaller architectures, suggesting that performance is limited by dataset characteristics rather than model size. Despite operating with a 1,024-token context window, four times larger than the limit used for the other models, Mistral shows no measurable gain, which may indicate that the cues discriminating reliable publishers are concentrated in the beginning of articles and further supports the hypothesis that increasing model size or input length alone cannot overcome limitations imposed by weak supervision. A key finding is that publisher reliability should not be modelled as a property of isolated articles: it emerges as a collective signal that becomes more stable when evidence is aggregated across a source, which supports a shift from article-centric formulations toward source-level inference.

Future research should extend this framework to multilingual settings, explore confidence-aware aggregation strategies, and develop training objectives tailored to ordinal classification. The value of such systems, however, will not be settled by accuracy alone. Whoever builds one has to show where its errors fall across the political spectrum, which is why we report that disparity alongside the scores.

Data and Code Availability

The evaluation protocol, the corpus definition and the material behind every claim are available at https://anonymous.4open.science/r/supplementary-material-DD54/. The repository holds the supplementary document; the `irgood` package, which implements the publisher-disjoint protocol, the tolerant metrics, the aggregation rule with its tie-breaking made explicit, and the bias audit as reusable code; the fold assignment for all 19,476 articles across 439 publishers; the aggregate statistics behind every figure and table; and a checklist of what a system of this kind has to report before anyone relies on it.

The NewsGuard reliability levels and leaning annotations are licensed to us and cannot be redistributed, so readers with a licence of their own can rebuild our exact splits from the released fold assignment, and readers without one can still see which corpus the claims rest on. We also do not publish our predicted reliability for named outlets: with a 31% publisher-level error rate that falls unevenly across the political spectrum, such a list would be the harm this paper argues against. That this line of work depends on proprietary ground truth is itself worth stating. Fine-tuned weights follow on acceptance, with a model card recording intended and out-of-scope uses alongside the error rates reported here.

Disclosure of Interests. The authors have no competing interests to declare.

References

1. Adji, S., de Lhoneux, M., Spinde, T.: HT-MAGPIE: a hierarchical transformer for article reliability classification. Datenbank-Spektrum **26**(1), 15–24 (2026). https://doi.org/10.1007/S13222-026-00526-Y, https://doi.org/10.1007/s13222-026-00526-y
2. Anonymous: Supplementary material for 'news publisher reliability from article collections: A publisher-disjoint evaluation' (2026), https://anonymous.4open.science/r/supplementary-material-DD54/supplementary.pdf, anonymised supplementary material accompanying this submission
3. Azizov, D., Mujahid, Z.M., AlQuabeh, H., Nakov, P., Liang, S.: SAFARI: Cross-lingual bias and factuality detection in news media and news articles. In: Findings of the Association for Computational Linguistics: EMNLP 2024. pp. 12217–12231. Association for Computational Linguistics (Nov 2024). https://doi.org/10.18653/v1/2024.findings-emnlp.712
4. Bazmi, P., Asadpour, M., Shakery, A.: Multi-view co-attention network for fake news detection by modeling topic-specific user and news source credibility. Information Processing and Management **60**(1) (2023)
5. Bohacek, M., Bravansky, M., Trhlík, F., Moravec, V.: Czech-ing the news: Article trustworthiness dataset for Czech. In: Barnes, J., De Clercq, O., Klinger, R. (eds.) Proceedings of the 13th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis. pp. 96–109. Association for Computational Linguistics, Toronto, Canada (Jul 2023). https://doi.org/10.18653/v1/2023.wassa-1.10, https://aclanthology.org/2023.wassa-1.10/
6. Burdisso, S., Sanchez-cortes, D., Villatoro-tello, E., Motlicek, P.: Reliability estimation of news media sources: Birds of a feather flock together. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers). pp. 6900–6918. Association for Computational Linguistics (2024). https://doi.org/10.18653/v1/2024.naacl-long.383
7. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). pp. 4171–4186 (2019)
8. Fogg, B., Tseng, H.: The elements of computer credibility. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. pp. 80–87. CHI '99, Association for Computing Machinery (1999). https://doi.org/10.1145/302979.303001
9. Kim, A., Moravec, P., Dennis, A.: Combating fake news on social media with source ratings: The effects of user and expert reputation ratings. Journal of Management Information Systems **36**(3), 931–968 (2019). https://doi.org/10.1080/07421222.2019.1628921
10. Leite, J., Razuvayevskaya, O., Bontcheva, K.e.a.: Weakly supervised veracity classification with llm-predicted credibility signals. EPJ Data Sci. **14**(15) (2025)
11. Lin, H., Lasser, J., Lewandowsky, S., Cole, R., Gully, A., Rand, D.G., Pennycook, G.: High level of correspondence across different news domain quality rating sets. PNAS nexus **2**(9), pgad286 (2023)

12. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. In: International Conference on Learning Representations (2020), https://openreview.net/forum?id=SyxS0T4tvS
13. Longo, A., Zappatore, M., Colella, D.D., Zampino, F., Calò, A.: xfair: A multi-layer approach to data fairness assessment and data fairification. ACM J. Data Inf. Qual. **17**(4), 25:1–25:29 (2025). https://doi.org/10.1145/3769113, https://doi.org/10.1145/3769113
14. Mauricio, P.G., Quesada-Lopez, C., Martinez, A., Jenkins, M.: Automatic classification of web news: A systematic mapping study. In: Intelligent Systems and Applications (IntelliSys 2020). pp. 558–574. Springer (2020). https://doi.org/10.1007/978-3-030-52249-0_44
15. Metzger, M., AJ, F.: Digital Media, Youth, and Credibility. MIT Press (2007 (1st edition))
16. Mistral AI: Mistral small 3. Model card: `mistralai/Mistral-Small-24B-Base-2501` (2025), https://mistral.ai/news/mistral-small-3/, 24B-parameter dense model
17. Monahov, A.: Improved accuracy metrics for classification with imbalanced data and where distance from the truth matters, with the wconf r package. Available at SSRN 4802336 (2024)
18. Nakov, P., An, J., Kwak, H., Manzoor, M.A., Mujahid, Z.M., Sencar, H.T.: A survey on predicting the factuality and the bias of news media. In: Findings of the Association for Computational Linguistics: ACL 2024. pp. 15947–15962 (2024). https://doi.org/10.18653/v1/2024.findings-acl.944
19. Pratelli, M., Petrocchi, M.: A structured analysis of journalistic evaluations for news source reliability. In: Workshop Proceedings of the 16th International AAAI Conference on Web and Social Media (2022), https://doi.org/10.36190/2022.51
20. Pratelli, M., Saracco, F., Petrocchi, M.: TROPIC - trustworthiness rating of online publishers through online interactions calculation. In: Hauff, C., Macdonald, C., Jannach, D., Kazai, G., Nardini, F.M., Pinelli, F., Silvestri, F., Tonellotto, N. (eds.) Advances in Information Retrieval - 47th European Conference on Information Retrieval, ECIR 2025, Lucca, Italy, April 6-10, 2025, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 15575, pp. 407–412. Springer (2025). https://doi.org/10.1007/978-3-031-88717-8_30, https://doi.org/10.1007/978-3-031-88717-8_30
21. Przybyla, P., Soto, A.: When classification accuracy is not enough: Explaining news credibility assessment. Information Processing and Management **58**(5), 102653 (2021). https://doi.org/https://doi.org/10.1016/j.ipm.2021.102653, https://www.sciencedirect.com/science/article/pii/S0306457321001412
22. Rogers, A., Kovaleva, O., Rumshisky, A.: A primer in bertology: What we know about how bert works. Transactions of the association for computational linguistics **8**, 842–866 (2021)
23. Spinde, T., Richter, E., Wessel, M., Kulshrestha, J., Donnay, K.: What do twitter comments tell about news article bias? assessing the impact of news article bias on its perception on twitter. Online Soc. Networks Media **37-38**, 100264 (2023). https://doi.org/10.1016/J.OSNEM.2023.100264, https://doi.org/10.1016/j.osnem.2023.100264
24. Srba, I., et al.: A survey on automatic credibility assessment using textual credibility signals in the era of large language models. ACM Trans. Intell. Syst. Technol. (Sep 2025). https://doi.org/10.1145/3770077, just Accepted
25. Team, L.: The llama 3 herd of models. CoRR **abs/2407.21783** (2024). https://doi.org/10.48550/ARXIV.2407.21783, https://doi.org/10.48550/arXiv.2407.21783

26. Verhoeven, I., Mishra, P., Shutova, E.: Yesterday's news: Benchmarking multi-dimensional out-of-distribution generalization of misinformation detection models. Computational Linguistics pp. 1–58 (2025)
27. Wardle, C., Derakhshan, H.: Information disorder: Toward an interdisciplinary framework for research and policy making. Tech. rep., Council of Europe (2017)
28. Williams, E.M., Carragher, P., Carley, K.M.: Bridging social media and search engines: Dredge words and the detection of unreliable domains. In: An, J., Lin, Y., Mejova, Y., Mustafaraj, E., Kulshrestha, J., Weber, I. (eds.) Proceedings of the Nineteenth International AAAI Conference on Web and Social Media, June 23-26, 2025, Copenhagen, Denmark. pp. 2030–2043. AAAI Press (2025). https://doi.org/10.1609/ICWSM.V19I1.35917, https://doi.org/10.1609/icwsm.v19i1.35917
29. Yang, K.C., Menczer, F.: Accuracy and political bias of news source credibility ratings by large language models. In: Proceedings of the 17th ACM Web Science Conference 2025. pp. 127–137. Websci '25, Association for Computing Machinery, New York, NY, USA (2025). https://doi.org/10.1145/3717867.3717903, https://doi.org/10.1145/3717867.3717903